

NT-C-004 | Data Analytics | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & Business Intelligence
Target learners	Learners targeting reporting, MIS, business-data and junior analyst roles.
Prerequisites	Basic computer use; spreadsheet familiarity is helpful.
Approved duration	Approved working range: 14-18 weeks / 160-220 guided hours.
Final outcome	Clean, analyse and present business data using spreadsheets, SQL, Python and BI dashboards.
Assessment profile	Data/AI

Approved tools / platforms

Excel, SQL, Python, pandas, Power BI, Git or shared portfolio repository.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-004 | Data Analytics | NT-C-004-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	Business questions, data types, KPIs; Functions, lookup, pivot, cleaning
Completion evidence	Problem statement and KPI map; Excel analysis workbook
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Analytics foundation

Objective: Business questions, data types, KPIs

Observable evidence: Problem statement and KPI map

Concept	Approved knowledge statement	Application behaviour	Evidence
Business questions	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when business questions is appropriate; apply it in a Data Analytics task; verify the result.	Problem statement and KPI map
data types	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when data types is appropriate; apply it in a Data Analytics task; verify the result.	Problem statement and KPI map
KPIs	A defined metric tied to a business objective and decision. In Data Analytics, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when kpis is appropriate; apply it in a Data Analytics task; verify the result.	Problem statement and KPI map

Module 2 - Excel analysis

Objective: Functions, lookup, pivot, cleaning

Observable evidence: Excel analysis workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
Functions	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when functions is appropriate; apply it in a Data Analytics task; verify the result.	Excel analysis workbook
lookup	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when lookup is appropriate; apply it in a Data Analytics task; verify the result.	Excel analysis workbook
pivot	An interactive aggregation used to summarise and explore tabular data. In Data Analytics, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when pivot is appropriate; apply it in a Data Analytics task; verify the result.	Excel analysis workbook
cleaning	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when cleaning is appropriate; apply it in a Data Analytics task; verify the result.	Excel analysis workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Problem statement and KPI map; Excel analysis workbook, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-004 | Data Analytics | NT-C-004-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Queries, joins, grouping, windows; Missing, duplicate, invalid and inconsistent data
Completion evidence	SQL case exercise; Cleaned dataset
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - SQL

Objective: Queries, joins, grouping, windows

Observable evidence: SQL case exercise

Concept	Approved knowledge statement	Application behaviour	Evidence
Queries	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when queries is appropriate; apply it in a Data Analytics task; verify the result.	SQL case exercise
joins	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when joins is appropriate; apply it in a Data Analytics task; verify the result.	SQL case exercise
grouping	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when grouping is appropriate; apply it in a Data Analytics task; verify the result.	SQL case exercise
windows	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when windows is appropriate; apply it in a Data Analytics task; verify the result.	SQL case exercise

Module 4 - Data quality

Objective: Missing, duplicate, invalid and inconsistent data

Observable evidence: Cleaned dataset

Concept	Approved knowledge statement	Application behaviour	Evidence
Missing	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when missing is appropriate; apply it in a Data Analytics task; verify the result.	Cleaned dataset
duplicate	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when duplicate is appropriate; apply it in a Data Analytics task; verify the result.	Cleaned dataset
invalid	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when invalid is appropriate; apply it in a Data Analytics task; verify the result.	Cleaned dataset
inconsistent data	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when inconsistent data is appropriate; apply it in a Data Analytics task; verify the result.	Cleaned dataset

Remediation: If the learner cannot explain the purpose, complete the stated task or provide SQL case exercise; Cleaned dataset, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-004 | Data Analytics | NT-C-004-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	pandas, transformation, automation; Summary, variation, correlation, interpretation
Completion evidence	Analysis notebook; Statistical insight note
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Python analytics

Objective: pandas, transformation, automation

Observable evidence: Analysis notebook

Concept	Approved knowledge statement	Application behaviour	Evidence
pandas	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when pandas is appropriate; apply it in a Data Analytics task; verify the result.	Analysis notebook
transformation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when transformation is appropriate; apply it in a Data Analytics task; verify the result.	Analysis notebook
automation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when automation is appropriate; apply it in a Data Analytics task; verify the result.	Analysis notebook

Module 6 - Statistics

Objective: Summary, variation, correlation, interpretation

Observable evidence: Statistical insight note

Concept	Approved knowledge statement	Application behaviour	Evidence
Summary	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when summary is appropriate; apply it in a Data Analytics task; verify the result.	Statistical insight note
variation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when variation is appropriate; apply it in a Data Analytics task; verify the result.	Statistical insight note
correlation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when correlation is appropriate; apply it in a Data Analytics task; verify the result.	Statistical insight note
interpretation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when interpretation is appropriate; apply it in a Data Analytics task; verify the result.	Statistical insight note

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Analysis notebook; Statistical insight note, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-004 | Data Analytics | NT-C-004-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Charts, dashboards, filters, storytelling; End-to-end analysis and presentation
Completion evidence	Interactive dashboard; Decision-ready report
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Visualization & BI

Objective: Charts, dashboards, filters, storytelling

Observable evidence: Interactive dashboard

Concept	Approved knowledge statement	Application behaviour	Evidence
Charts	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when charts is appropriate; apply it in a Data Analytics task; verify the result.	Interactive dashboard
dashboards	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dashboards is appropriate; apply it in a Data Analytics task; verify the result.	Interactive dashboard
filters	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when filters is appropriate; apply it in a Data Analytics task; verify the result.	Interactive dashboard
storytelling	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when storytelling is appropriate; apply it in a Data Analytics task; verify the result.	Interactive dashboard

Module 8 - Business capstone

Objective: End-to-end analysis and presentation

Observable evidence: Decision-ready report

Concept	Approved knowledge statement	Application behaviour	Evidence
End-to-end analysis	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when end-to-end analysis is appropriate; apply it in a Data Analytics task; verify the result.	Decision-ready report
presentation	A defined competency within Data Analytics. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when presentation is appropriate; apply it in a Data Analytics task; verify the result.	Decision-ready report

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Interactive dashboard; Decision-ready report, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-004 | Data Analytics | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-004-SK01	Analytics foundation	Prerequisite foundation	Problem statement and KPI map	Critical
NT-C-004-SK02	Excel analysis	M01 plus earlier foundations	Excel analysis workbook	Critical
NT-C-004-SK03	SQL	M02 plus earlier foundations	SQL case exercise	Supporting
NT-C-004-SK04	Data quality	M03 plus earlier foundations	Cleaned dataset	Supporting
NT-C-004-SK05	Python analytics	M04 plus earlier foundations	Analysis notebook	Supporting
NT-C-004-SK06	Statistics	M05 plus earlier foundations	Statistical insight note	Critical
NT-C-004-SK07	Visualization & BI	M06 plus earlier foundations	Interactive dashboard	Supporting
NT-C-004-SK08	Business capstone	M07 plus earlier foundations	Decision-ready report	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-004 | Data Analytics | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-004-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Analytics foundation and Excel analysis?	The learner should provide Problem statement and KPI map; Excel analysis workbook and explain how the work applies Business questions, data types, KPIs; Functions, lookup, pivot, cleaning. Recall alone is insufficient.
NT-C-004-S01-EX02	Scenario/remediation	A learner reports completing Excel analysis but cannot produce Excel analysis workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-004-S01, request Excel analysis workbook, and allow a new equivalent test only after remediation.
NT-C-004-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of SQL and Data quality?	The learner should provide SQL case exercise; Cleaned dataset and explain how the work applies Queries, joins, grouping, windows; Missing, duplicate, invalid and inconsistent data. Recall alone is insufficient.
NT-C-004-S02-EX02	Scenario/remediation	A learner reports completing Data quality but cannot produce Cleaned dataset. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-004-S02, request Cleaned dataset, and allow a new equivalent test only after remediation.
NT-C-004-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Python analytics and Statistics?	The learner should provide Analysis notebook; Statistical insight note and explain how the work applies pandas, transformation, automation; Summary, variation, correlation, interpretation. Recall alone is insufficient.
NT-C-004-S03-EX02	Scenario/remediation	A learner reports completing Statistics but cannot produce Statistical insight note. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-004-S03, request Statistical insight note, and allow a new equivalent test only after remediation.
NT-C-004-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Visualization & BI and Business capstone?	The learner should provide Interactive dashboard; Decision-ready report and explain how the work applies Charts, dashboards, filters, storytelling; End-to-end analysis and presentation. Recall alone is insufficient.
NT-C-004-S04-EX02	Scenario/remediation	A learner reports completing Business capstone but cannot produce Decision-ready report. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-004-S04, request Decision-ready report, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-004 | Data Analytics | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Sales performance analysis	Excel, KPIs, pivots	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	Lead conversion analysis	SQL, cleaning, dashboard	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Branch operations analytics	Python/SQL, BI, presentation	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-004 | Data Analytics | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	Junior Data Analyst	Excel 25% [critical]; SQL 25% [critical]; cleaning 25%; dashboards 25%	Analysis case and dashboard	If a critical skill is below 60 or analysis case and dashboard is missing, return Improve First or Needs Evidence with the linked module/project.
2	Reporting Analyst	KPIs 34% [critical]; reporting 33% [critical]; visualization 33%	Automated report	If a critical skill is below 60 or automated report is missing, return Improve First or Needs Evidence with the linked module/project.
3	MIS / BI Analyst - Junior	Excel 34% [critical]; SQL 33% [critical]; Power BI 33%	Management dashboard	If a critical skill is below 60 or management dashboard is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-004 | Data Analytics | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Data Analytics for?	Learners targeting reporting, MIS, business-data and junior analyst roles.
What should I know before starting?	Basic computer use; spreadsheet familiarity is helpful.
How long is the approved learning journey?	Approved working range: 14-18 weeks / 160-220 guided hours.
What will I be able to demonstrate?	Clean, analyse and present business data using spreadsheets, SQL, Python and BI dashboards.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Data Analytics knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-004; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```